

Practice Assignment

Course: Python for Data Science

Problem Statement

You will be given a dataset stored in a **text file (.txt)** containing:

- Two numerical features
- One categorical target variable (string)

Your task is to implement the complete machine learning pipeline using **Object-Oriented Programming (OOP)** and to build a **classification model from scratch** using **NumPy only** (for model creation).

Important Constraints:

- **Do NOT use sklearn models**
- NumPy is allowed for mathematical operations
- sklearn is allowed **only** for preprocessing and evaluation

Q1. File Handling and Data Loading

- a) Read the dataset from a text file using Python file handling.
- b) Store the numerical features and categorical target separately.
- c) Handle malformed or missing rows safely.

OOP Requirement: Create a class `DataLoader` with appropriate attributes and methods.

Q2. Exploratory Data Analysis (EDA)

Before building any machine learning model, perform exploratory analysis to understand the dataset.

- a) Compute and report:
 - Mean, median, and standard deviation of each numerical feature
 - Minimum and maximum values of each feature
- b) Identify whether the dataset is **balanced or imbalanced** with respect to the target classes. Support your answer using class counts.
- c) Analyze the relationship between features and target:

- Do higher values of features tend to favor a particular class?
 - Provide reasoning based on summary statistics.
- d) Detect the presence of potential outliers using statistical reasoning (e.g., unusually high or low values).

Note: Focus on **analysis and interpretation**. Plots are optional.

Q3. Data Preprocessing

Perform the following preprocessing steps:

- Encode the categorical target variable into numerical form.
- Normalize the numerical features.
- Split the dataset into training and testing sets (80:20).

Note: You may use `sklearn.preprocessing` for this question.

OOP Requirement: Create a class `Preprocessor` to handle preprocessing tasks.

Q4. Model Implementation from Scratch

Implement a **classification model from scratch** using NumPy. You may choose **ONE** of the following:

- Logistic Regression
- A simple two-layer Neural Network

The implementation must include:

- Parameter initialization
- Forward propagation
- Loss computation
- Gradient computation
- Gradient descent optimization

OOP Requirement: Define a model class with methods such as:

- `fit()`
- `predict()`

Q5. Model Evaluation

Evaluate your model on the test data using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

Task:

- Report all metrics before parameter tuning.

OOP Requirement: Create a class `Evaluator` to compute evaluation metrics.

Q6. Parameter Tuning and Performance Improvement

- Tune at least **two hyperparameters** (e.g., learning rate, number of epochs).
- Retrain the model using the tuned parameters.
- Re-evaluate the model using all four metrics.

Comparison Table:

Metric	Before Tuning	After Tuning
Accuracy		
Precision		
Recall		
F1-score		

Q7. Conceptual and Reflection Questions

Answer the following questions briefly and clearly:

- Why is feature scaling important before applying gradient descent?
- What is the role of the learning rate in model training? What happens if it is too large or too small?
- Why can accuracy be misleading for imbalanced datasets? Which metric would you trust more and why?
- Explain the advantage of implementing a model from scratch instead of directly using library functions.
- How does Object-Oriented Programming improve the readability and maintainability of machine learning code?

General Instructions

- Follow proper OOP principles (encapsulation and modularity).
- Clearly comment your code.
- Use meaningful variable and class names.